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Order no.: 63193794
Order received: 16 Feb. 2024
Sample type / Sample collection date:
blood, CentoCard® / 28 Jan. 2024
Report date: 10 Apr. 2024
Report type: Final report



Patient no: **1886117**, First Name: **Faisal Raed Rajab**, Last Name: **Alzahrani**
DOB: **31 Dec. 2023**, Sex: **male**, Your ref.: **8100371892**

Additional report recipient(s): Dr. Hanem Sayed Ahmed Abdelraouf, Prince Sultan Military Medical City (PSMMC), Clinical Genetics, King Abdulaziz, 11159 Riyadh, Saudi Arabia; Prof. Dr. Amal Alhashem, Doctor's office of Dr. Amal Al Hashem, Makkah Road, 11159 Riyadh, Saudi Arabia

Test(s) requested: CentoXome® MOx 1.0 Solo

CLINICAL INFORMATION

Abnormal left ventricle morphology; Abnormal left ventricular function; Abnormal morphology of left ventricular trabeculae; Hydronephrosis; Left ventricular dilatation; Mildly reduced left ventricular ejection fraction; Mitral regurgitation; Patent foramen ovale; Reduced left ventricular ejection fraction; Tricuspid regurgitation

(Clinical information indicated above follows HPO nomenclature.)

Family history: Yes.

Brother: Hypoplastic left heart, Neonatal death.

Consanguineous parents: Yes.



POTENTIALLY RELEVANT RESULT
Pathogenic variant identified

INTERPRETATION

A pathogenic variant was identified in heteroplasmy (40%) in the mitochondrial *MT-TL1* gene. This variant is the most frequent variant associated with mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes (MELAS), and is associated with diverse clinical manifestations (i.e., progressive external ophthalmoplegia, diabetes mellitus, cardiomyopathy, deafness) that collectively constitute a wide spectrum ranging from MELAS at the severe end to asymptomatic carrier status.

No further clinically relevant variants related to the described phenotype were detected.

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RECOMMENDATIONS

- Clinical evaluation to assess the phenotypic overlap with the detected variant is recommended.
- As the percentage of heteroplasmy of pathogenic mtDNA variants may vary between organs and tissues in an individual, it is recommended to study other tissues in this patient (e.g., saliva, urine) for better integration of this result.
- NGS-based targeted analysis is recommended for the mother (in different tissues) to determine segregation. NGS-based targeted analysis is also recommended for other affected or at-risk matrilineal relatives.
- Specialized clinical follow-up and surveillance for possible additional manifestations associated with the disease is recommended.
- If the detected variant is not considered to contribute to or fully explain the patient's phenotype, reevaluation of the sequence dataset is recommended every 12 months or if there are phenotypic changes. Additionally, proceeding to genome sequencing should be considered.
- Genetic counselling is recommended.

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MAIN FINDINGS

SEQUENCE VARIANTS							
GENE	VARIANT COORDINATES	AMINO ACID CHANGE	SNP IDENTIFIER	ZYGOSITY	IN SILICO PARAMETER S*	ALLELE FREQUENCY ES**	TYPE AND CLASSIFICATION***
MT-TL1	NC_012920.1:m.3243A>G	-		Heteroplasmy (40% of 2104 NGS reads)	MitoTIP: Pathogenic HmtVar: Pathogenic	gnomAD: - MitoMap: 0.019%	Alteration Pathogenic (class 1)

Variant annotation based on CentoCloud Bioinformatics pipeline. * AlignGVD: C0: least likely to interfere with function, C65: most likely to interfere with function; splicing predictions: Ada and RF scores. ** Genome Aggregation Database (gnomAD), Exome Sequencing Project (ESP), 1000Genome project (1000G) and CentoMD® (latest database available). *** based on ACMG recommendations.

VARIANT INTERPRETATION

MT-TL1, m.3243A>G -

A heteroplasmic variant, m.3243A>G (40% heteroplasmy), was identified in the *MT-TL1* gene encoding the mitochondrial tRNA^{Leu}(UUR). This variant has been previously reported as disease-causing for MELAS (mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes), and it is the most common pathogenic variant, present in about 80% of individuals with typical clinical findings (PMID: 20301411). It is annotated in MitoMap. This variant has been associated with diverse clinical manifestations (i.e., progressive external ophthalmoplegia, diabetes mellitus, cardiomyopathy, deafness) that collectively constitute a wide spectrum ranging from MELAS at the severe end to asymptomatic carrier status. More severe phenotypes may be the result of a higher abundance of the pathogenic variant in affected organs. It has also been identified in patients with autism spectrum disorder (PMID: 35778412, 21263444). It is classified as pathogenic according to the recommendations of CENTOGENE implementation of the ACMG/AMP/ClinGen SVI guidelines.

Pathogenic variants in the *MT-TL1* gene are associated with several mitochondrial disorders, including MELAS syndrome, MERRF, diabetes with maternally transmitted deafness, and Leigh syndrome. MELAS syndrome, comprising mitochondrial myopathy, encephalopathy, lactic acidosis, and stroke-like episodes, is a genetically heterogeneous mitochondrial disorder with a variable clinical phenotype. The disorder is accompanied by features of central nervous system involvement, including seizures, hemiparesis, hemianopsia, cortical blindness, and episodic vomiting (OMIM® 540000).

The m.3243A>G variant specifically is the one most frequently associated with MELAS syndrome. It is associated with various clinical manifestations (i.e. progressive external ophthalmoplegia, diabetes mellitus, cardiomyopathy, deafness) that together constitute a broad spectrum ranging from MELAS in the severe extreme to asymptomatic carrier state. The more severe phenotypes may be the result of a higher abundance of the variant in the affected organs (PMID: 20301411).

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SECONDARY FINDINGS

If consent is provided, in line with ACMG recommendations (ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing; Genetics in Medicine, 2023; PMID: 37347242) we report secondary findings, i.e. relevant pathogenic and likely pathogenic variants in the recommended genes for the indicated phenotypes in this publication.

We did not detect any relevant variants in the genes for which secondary findings are reported.

CARRIERSHIP FINDINGS

In this table we list sequence variants previously ascertained or evaluated and classified in CENTOGENE as "pathogenic" and "likely pathogenic", in selected genes associated with recessive severe and early-onset Mendelian diseases. As only in-house classified variants are presented, it should not be considered a comprehensive list of variants in these genes and does not provide a complete list of potentially relevant genetic variants in the patient. The complete gene list can be found at www.centogene.com/carriership-findings (please contact CENTOGENE customer support if the gene list has been updated after this report was issued). Orthogonal validation was not performed for these variants. Therefore, if any variant is used for clinical management of the patient, confirmation by another method needs to be considered. Furthermore, the classification of these variants may change over time, however reclassification reports for these variants will not be issued. CENTOGENE is not liable for any missing variant in this list and/or any provided classification of the variants at a certain point of time. As the identified variants may indicate (additional) genetic risks or diagnoses in the patient and/or family and/or inform about reproductive risks, we recommend discussing these findings in the context of genetic counselling.

In accordance with this disclaimer no relevant pathogenic or likely pathogenic variant was identified.

CENTOGENE VARIANT CLASSIFICATION (BASED ON ACMG RECOMMENDATIONS)

Class 1 – Pathogenic

Class 2 – Likely pathogenic

Class 3 – Variant of uncertain significance (VUS)

Class 4 – Likely benign

Class 5 – Benign

Additionally, other types of clinically relevant variants can be identified (e.g. risk factors, modifiers).

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METHODS

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Genomic DNA is enzymatically fragmented, and target regions are enriched using DNA capture probes. These regions include approximately 41 Mb of the human coding exome (targeting > 98% of the coding RefSeq from the human genome build GRCh37/hg19), as well as the mitochondrial genome. The generated library is sequenced on an Illumina platform to obtain at least 20x coverage depth for > 98% of the targeted bases. An in-house bioinformatics pipeline, including read alignment to GRCh37/hg19 genome assembly and revised Cambridge Reference Sequence (rCRS) of the Human Mitochondrial DNA (NC_012920), variant calling, annotation, and comprehensive variant filtering is applied. All variants with minor allele frequency (MAF) of less than 1% in gnomAD database, and disease-causing variants reported in HGMD®, in ClinVar or in CentoMD® are evaluated. The investigation for relevant variants is focused on coding exons and flanking +/-10 intronic nucleotides of genes with clear gene-phenotype evidence (based on OMIM® information). All potential patterns for mode of inheritance are considered. In addition, provided family history and clinical information are used to evaluate identified variants with respect to their pathogenicity and disease causality. Variants are categorized into five classes (pathogenic, likely pathogenic, VUS, likely benign, and benign) along ACMG guidelines for classification of variants. All relevant variants related to the phenotype of the patient are reported. For CentoXome® MOx, if applicable, biochemical analysis is performed upon detection of relevant variants by sequencing. This enhances the diagnosis of metabolic disorders, optimizes variant classification, and helps to ascertain the eventual contribution to the phenotype; the list of enzyme-activity assays and biomarkers can be obtained at www.centogene.com/mox. CENTOGENE has established stringent quality criteria and validation processes for variants detected by NGS. Variants with low sequencing quality and/or unclear zygosity are confirmed by orthogonal methods. Consequently, a specificity of > 99.9% for all reported variants is warranted. Mitochondrial variants are reported for heteroplasmy levels of 15% or higher. The copy number variation (CNV) detection software has a sensitivity of more than 95% for all homozygous/hemizygous and mitochondrial deletions, as well as heterozygous deletions/duplications and homozygous/hemizygous duplications spanning at least three consecutive exons. For the uniparental disomy (UPD) screening, a specific algorithm is used to assess the well-known clinically relevant chromosomal regions (6q24, 7, 11p15.5, 14q32, 15q11q13, 20q13 and 20).

ANALYSIS STATISTICS

CentoXome® MOx 1.0 Solo

Targeted nucleotides covered	≥ 20x	98.98%
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LIMITATIONS

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The genetic results are interpreted in the context of the provided clinical findings, family history, and other laboratory data. Only variants in genes potentially related to the proband's medical condition are reported. Misinterpretation of results may occur if the provided genetic data or patient information is inaccurate and/or incomplete. If the obtained genetic results are not compatible with the clinical findings, additional testing should be considered.

The genes with mapping issues in GRCh37/hg19 genome assembly, the non-protein-coding disease-associated genes, and approximately 0.2 Mb of genomic regions that are hard to sequence by current enrichment technology and are without evidenced relevance for monogenic disorders, are excluded from this analysis. More complex genetic events such as inversions, translocations, and repeat expansions, are not analyzed in this test. The UPD detection is a screening method, and therefore false-positive and false-negative results may occur. In addition, due to technology limitations, certain regions may be poorly covered, or not covered at all. In these regions and others encompassing repetitive, high-homology (such as pseudogene homology), and GC-rich sequences, relevant variants can be missed. Extremely low-coverage calls (homo/hemizygous or heterozygous calls with less than three or four reads, respectively) are expected to be artifacts based on our extensive validations and are consequently not considered during the analysis. Heterozygous CNVs spanning less than three exons cannot reliably be detected, are therefore excluded from routine analysis, and will only be inspected and reported upon medical or technical indication. The CNV detection sensitivity is decreased for repetitive and homologous regions, such as pseudogenes. Mitochondrial variants with heteroplasmy levels below 15% may not be detected. It is expected that lower quality samples (prenatal, product of conception, blood from patients with hematologic disorders, and highly degraded DNA) may generate lower quality NGS data; in these cases, CNV analysis and/or mitochondrial genome analysis may not be possible to perform. Potential aberrant splicing is assessed with splice prediction tools. Intronic variants that are beyond 10 nucleotides from exon-intron boundaries are not considered for aberrant splicing analysis, with the exception of known pathogenic splicing variants evidenced by external sources.

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ADDITIONAL INFORMATION

This test was developed, and its performance was validated, by CENTOGENE. The US Food and Drug Administration (FDA) has determined that clearance or approval of this method is not necessary and thus neither have been obtained. This test has been developed for clinical purposes. All test results are reviewed, interpreted and reported by our scientific and medical experts.

To exclude mistaken identity in your clinic, several guidelines recommend testing a second sample that is independently obtained from the proband. Please note that any further analysis will result in additional costs.

The classification of variants can change over the time. Please feel free to contact CENTOGENE (customer.support@centogene.com) in the future to determine if there have been any changes in classification of any reported variants.

DISCLAIMER

Any preparation and processing of a sample from patient material provided to CENTOGENE by a physician, clinical institute or a laboratory (by a "Partner") and the requested genetic and/or biochemical testing itself is based on the highest and most current scientific and analytical standards. However, in very few cases genetic or biochemical tests may not show the correct result, e.g. because of the quality of the material provided by a Partner to CENTOGENE or in cases where any test provided by CENTOGENE fails for unforeseeable or unknown reasons that cannot be influenced by CENTOGENE in advance. In such cases, CENTOGENE shall not be responsible and/or liable for the incomplete, potentially misleading or even wrong result of any testing if such issue could not be recognized by CENTOGENE in advance.

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